

Multiplayer online battle arena

Multiplayer online battle arena (MOBA)^[a] is a subgenre of strategy video games in which two teams of players compete on a structured battlefield, each controlling a single character with distinctive abilities that grow stronger as the match progresses.^[1] The objective is to destroy the enemy team's main structure while defending one's own. In some MOBA games, the objective can be defeating every player on the enemy team. Matches emphasize team coordination, tactical choices, and real-time combat. Players are assisted by computer-controlled units that periodically spawn in groups and march along set paths toward their enemy's base, which is heavily guarded by defensive structures. Players can influence these units by eliminating enemy waves or supporting their own, affecting lane control and map pressure. This type of multiplayer online video games originated as a subgenre of real-time strategy (RTS); however, most of the traditional RTS elements, such as building construction and unit production, were removed in favor of a more focused player-versus-player experience. The genre blends elements of real-time strategy, role-playing, and action games, combining strategic depth with individual character progression and fast-paced combat.

The first widely accepted game in the genre was *Aeon of Strife* (AoS), a fan-made custom map released in 2002 for *StarCraft*, in which four players each control a single powerful unit and, aided by weak computer-controlled units, compete against a stronger computer.^{[2][3]} *Defense of the Ancients* (*DotA*) was created in 2003 by the *Warcraft III* modding community for *Warcraft III: Reign of Chaos* and its expansion, *The Frozen Throne*, with a map based on AoS. *DotA* was one of the first major titles to establish the core mechanics of the MOBA genre, serving as a direct inspiration for later titles, and the first MOBA for which sponsored tournaments were held.^[4] It was followed by two spiritual successors, *League of Legends* (2009) and *Heroes of Newerth* (2010), a standalone sequel, *Dota 2* (2013), and other games in the genre, including *Smite* (2014) and *Heroes of the Storm* (2015).^{[5][6]}

Through the years, the MOBA genre has played a significant role in the rise of competitive esports. By the early 2010s, the genre had established itself as a major component of the esports landscape, with prize pools reaching over US\$60 million in 2018, accounting for 40% of the total esports prize pools that year. Major esports professional tournaments are held in venues that can hold tens of thousands of spectators and are streamed online. A strong fanbase has opened up the opportunity for sponsorship and advertising, eventually leading the genre to become a global cultural phenomenon.^[7]

Gameplay

Each match starts with two opposing teams, typically with five players each. Players work together as a team to achieve the ultimate victory condition, which is to destroy their enemy's base whilst protecting their own.^[8] Both teams usually have their main structures placed on opposite sides of the battlefield. The first team to destroy the opponents' main structure wins the match,^[8] though some games have the option of different victory conditions.^[9] Destroying other structures in the enemy's base may provide other benefits. Defensive structures, usually automatic "towers", are in place to prevent this. Each team is assisted by relatively weak computer-controlled units, called "minions", that periodically spawn in groups at both bases, marching down predefined paths (called "lanes") toward the enemy base. While minions

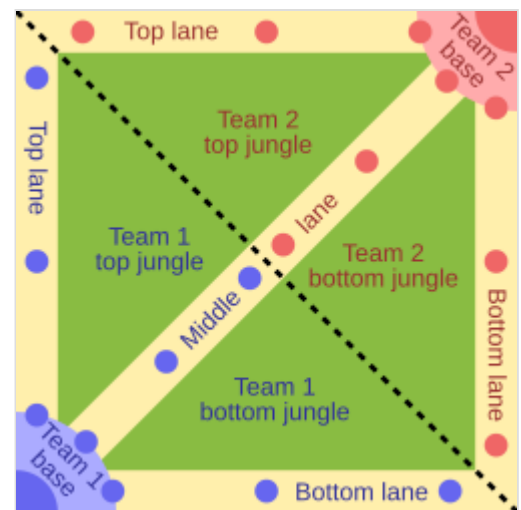
naturally engage opposing forces, players can support them to increase their effectiveness in pushing through enemy defenses.^{[10][11]} There are typically three "lanes" on the battlefield that serve as primary paths between bases. The lanes are known as top, middle and bottom lane, or, in gamer shorthand – "top", "mid" and "bot". Between the lanes is an uncharted area called "jungle".^[12] The "jungle" is home to neutral monsters hostile to both teams and appear in marked locations on the map known as "camps".^[13] Defeating these monsters grants various benefits to the players and their team, such as growth in power, temporary buffs, or assistance in pushing the lane. Effective control over lanes and jungle objectives is crucial for maintaining map pressure and dictating the flow of the match.^{[14][15]}

The games are usually played on a battlefield shown from an isometric perspective, but certain MOBAs are played from a third-person or side-view perspective.^[3] The battlefield is represented in the interface by the mini-map.^[16]

A player controls a single powerful in-game unit, known as a "hero" or "champion," with a distinctive set of abilities and a unique playstyle.^[1] Heroes gain experience points and gold by being near defeated enemies or delivering the killing blow. Experience allows them to level up and enhance abilities, while gold is used to purchase items that improve their power. Most heroes have four abilities that can be upgraded as they progress. If a hero runs out of health points and dies, they are removed from active play until a respawn timer counts down to zero, at which point the hero respawns in their base. The respawn time increases as the game progresses and heroes level up.^[17]

Heroes typically fall into one of several roles, such as tank, damage dealer, and support, each with unique designs, strengths, and weaknesses.^[1] MOBAs typically offer a large number of viable playable heroes – *League of Legends*, for instance, began with 40, and added characters over time, reaching 100 in 2012 and 150 in 2020.^{[18][19][20]} This adds to the overall complexity of the game as players must be aware of an increasing list of available characters. Choosing the right character is a skill, requiring players to evaluate available options and select one that fits their skill set, team composition, and opposing picks.^[21] Players often find a hero they excel at, referred to as a "main", and familiarize themselves with the remaining roster.^[22] Each hero is limited in the roles they can fulfill. No single hero is supposed to be powerful enough to win the game without team support. This creates a strong emphasis on teamwork and cooperation.^[23] The genre rewards players that are capable of cooperating with teammates to execute an effective strategy, enabling full potential of their individual abilities and mechanical skills.^[24]

Each player typically receives a small amount of gold per second during the course of the game. Moderate amounts of gold are rewarded for killing hostile computer-controlled units and larger amounts are rewarded for killing enemy heroes. Gold is used by heroes to buy a variety of different items that range in price and impact. For the most part, this involves improving the combat viability of the hero, although there may be other items that support the hero or team as a whole in different ways.^[25] As the heroes of each team get stronger, they can use multiple strategies to gain an advantage. These strategies



A generic map layout popular in MOBA games. Yellow lines are the "lanes" where the action is focused. Blue and red dots are the "towers", the main line of defense for both teams. Green area is the jungle. Two light-colored areas are the teams' bases which encompass the blue and red corners, the structures upon which destruction results in victory.

can include securing objectives, killing enemy heroes, and gaining levels by defeating computer-controlled units. As a team grows stronger, they gain greater control over the map, apply more pressure on the opponent, and become more capable of dismantling the enemy's defenses and structures, ultimately leading to the destruction of their base.^[21]

Character classes and roles

In most MOBAs, playable characters have assigned classes such as "tank", "bruiser", "marksman", "mage", "fighter", "assassin", "support" and "healer", with each classification denoting various different skill sets and proficiencies.^{[26][27]} During the match, characters can be played in roles such as "carry", "support" and "ganker"; however, the number and type of roles can differ depending on the game.^{[28][29]} The carry role is expected to scale and itemize themselves to do the most damage against enemy characters and objectives, but may also require protection and assistance from their team members.^[28] Supports assist their team with abilities that aid allies and disable enemies, rather than dealing damage directly. Some supports have healing abilities which can be vital factor in the team composition's success, giving health and sustenance to their allies while limiting the enemy's options in terms of play patterns.^[30] Ganker roles are flexible, as they have both carry and support skills that are used to disrupt and eliminate enemies, thus giving their teammates an advantage over their opponents.^[28] Gankers can "act as a strategist, decision-maker or supporter depending on the team's needs."^[28] Player roles can be classified by the particular lane they are focusing on, such as "top laner", "mid laner", and "bottom laner",^[26] or by their role in a teamfight, such as "frontliner", "damage dealer", "healer", "flex", and the "offlaner".^[30]

Resemblance to other genres

As a fusion of real-time strategy (RTS), role-playing, and action games, MOBAs have many elements of established genres while still offering unique gameplay. In general, the design philosophy of the MOBA genre has moved away from constructing structures, army building, and controlling additional units in favor of hero-centric gameplay.^[24] However, some MOBA games have certain heroes that control a few specialized units,^[31] but not on the massive scale commonly found in RTS games.^[24] Much like real-time strategy games, structures and base defense play a crucial role in MOBAs, with the victory condition often being the destruction of the main structure in the enemy base. Players can find various friendly and enemy units on the map assisting each team. However, these units are computer-controlled and players usually do not control their movement or creation. Instead of building them, players rely on these units to move along the map's lanes.

Many defining elements of the action genre^{[32][33]} are represented in MOBA games. Players with better mechanical skills and quick reaction times typically excel relative to their peers. MOBAs often have a strong focus on micromanagement, involving mechanical abilities such as positioning, dodging, use of combo attacks, kiting, prediction and target selection.^[24] Direction-targeted abilities, or "skillshots", require precise aim and good timing in order to hit an enemy.^[34]



The Halcyon Fold map from the game *Vainglory* features a single lane connecting the two team bases, and the "jungle" underbrush beneath the lane.

The MOBA genre resembles role-playing games (RPGs) in gameplay, though the MOBA genre focuses on the multiplayer battle in the arena-like environment, whereas RPGs typically revolve around a single-player story and exploration of different locations.^{[35][33]} Some key features of MOBAs, such as control over one specific character in a party, growth in power over time, learning new thematic abilities, leveling and accumulation of experience points,^[36] usage of the mana resource,^[37] equipment and inventory management,^[38] completing quests,^[39] and fighting with powerful boss monsters,^{[14][15]} are also typical of role-playing games.

History

Origins

The 1989 Mega Drive/Genesis game *Herzog Zwei* has variously been cited either as a precursor to,^{[40][41]} or an early example of, the MOBA genre.^{[42][43]} It uses a similar formula, where each player controls a single command unit in one of two opposing sides on a battlefield.^{[41][43][40]} *Herzog Zwei*'s influence is apparent in several later MOBA games such as *Guilty Gear 2: Overture* (2007)^[44] and *AirMech* (2012).^[42] *Herzog Zwei* was also cited as an inspiration to the developers of *Warcraft* and *Starcraft*.^{[45][46]}

1998's *Future Cop: LAPD* has a strategic "Precinct Assault" mode similar to *Herzog Zwei* in which players can actively fight alongside generated non-player units.^{[47][48]} This could be regarded as the first example of MOBA gameplay, depending on the definition of the genre.^[49] The Windows version of *Future Cop: LAPD* allows online competitive play.^{[50][51]}

In the same year, Blizzard Entertainment released its best-selling real-time strategy game *StarCraft* (1998) with a suite of game editing tools called *StarEdit*. These tools allowed players to design and create custom maps with non-standard rules and gameplay. A modder known as Aeon64 made a custom map named *Aeon of Strife* (AoS) that became popular. Some of the key features introduced in AoS became the foundation of the newborn genre.^{[3][52]} In the *Aeon of Strife* map, players controlled a single powerful hero unit fighting along three lanes which were protected by defensive towers. The terrain outside these lanes was nearly vacant. In early versions of the game, hero units did not have any particular special abilities. Instead, players spent gold on weapon and armor upgrades.^{[52][53]}

Establishing the genre: 2000s

In 2002, Blizzard released *Warcraft III: Reign of Chaos* (WC3), with the accompanying *Warcraft III World Editor*. Both the multiplayer online battle arena and tower defense subgenres took substantive shape within the WC3 modding community. A modder named Eul began converting *Aeon of Strife* into the *Warcraft III* engine, calling the map *Defense of the Ancients* (*DotA*). Eul substantially improved the complexity of play from the original *Aeon of Strife* mod. Shortly after creating the custom *DotA* map, Eul left the modding scene. With no clear successor, *Warcraft III* modders created a variety of maps based on *DotA* and featuring different heroes. In 2003, after the release of *WarCraft III: The Frozen Throne*, a map creator named Meian^[2] created a *DotA* variant closely modeled on Eul's map, but combining heroes from the many other versions of *DotA* that existed at the time. Called *DotA: Allstars*, it was inherited after a few months by a modder called Steve "Guinsoo" Feak, and under his guidance it became the dominant

map of the genre. After more than a year of maintaining the *DotA: Allstars* map, with the impending release of an update that significantly changed the map layout, Guinsoo left the development to his adjutant Neichus in the year 2005.^[2] After some weeks of development and some versions released, the latter turned over responsibility to a modder named IceFrog, who initiated large changes to the mechanics that deepened its complexity and capacity for innovative gameplay. The changes conducted by IceFrog were well-received and the number of users on the *Dota: Allstars* forum is thought to have peaked at over one million.^[53] *DotA* is widely regarded as the most significant influence on the MOBA genre, shaping its core mechanics and inspiring the development of numerous titles in the years to come.^[3]

Mainstream popularity: 2008–present

By 2008, the popularity of *DotA* had attracted commercial attention.^[54] Since the format was tied to the *Warcraft* property, developers began to work on their own "DOTA-style" video games.^{[55][56][57]} A Flash web game, named *Minions*, was created by The Casual Collective in 2008.^[58] Gas Powered Games released the first stand-alone commercial title in the genre, *Demigod* (2009).^[59] In late 2009, Riot Games' debut title *League of Legends* was released. It was initially designed by Steve Feak, one of the original creators of *DotA: Allstars*, who went on to apply many of the mechanics and lessons he learned from the mod.^{[60][61]} Riot began to refer to the game's genre as a multiplayer online battle arena (MOBA).^[62] Also in 2009, IceFrog, who had continued to develop *DotA: Allstars*, was hired by Valve, in order to design a sequel to the original map.^[53]

In 2010, S2 Games released *Heroes of Newerth*, with a large portion of its gameplay and aesthetics based on *DotA: Allstars*.^{[63][64]} The same year, Valve announced *Dota 2* and subsequently secured the franchise's intellectual property rights^{[65][66]} after being contested by Riot Games for the *DotA* trademark.^[67] In 2012, Activision Blizzard settled a trademark dispute with Valve over the usage of the *DOTA* name and announced their own standalone game which was eventually named *Heroes of the Storm*.^{[68][69][70][71][72]} *Dota 2* was released in 2013, and was referred to by Valve as an "action real-time strategy" game.^[62] In 2014, Hi-Rez Studios released *Smite*, a MOBA with a third-person perspective.^[73] *Heroes of the Storm* was released in 2015, featuring hero characters from *Warcraft III* and other Blizzard's franchises.^{[74][75]} Blizzard adopted their own personal dictation for their game's genre with "hero brawler", citing its focus on action.^[76]

With the expansion of the smartphone market, numerous MOBA titles have been released for portable devices,^[77] such as *Vainglory* (2014), and *Honor of Kings* (2015). An international adaptation of *Honor of Kings* developed by TiMi Studios and published by Tencent Games for markets outside mainland China, rebranded as *Arena of Valor* (2016), was released in the western market in 2017.^[78] In 2021, the Pokémon series released its first MOBA game in *Pokémon Unite*.^[79]

Next-generation wave and market saturation

During the last half of the 2010s, video game developers and publishers, following the success of *League of Legends* and *Dota 2*, tried to be part of the next-generation MOBA wave by putting their own twist in the genre,^{[80][81][82][83][84]} releasing games such as *Battlerite* (2017), and *AirMech* (2018). After years of development, many games which were supported by large publishers have not been fully released or their servers were shut down shortly after release. The most notable examples are *Dawngate* (2015)^[b] by

Electronic Arts,^[85] DC Comics-based *Infinite Crisis* (2015) by Warner Bros.,^[86] *Arena of Fate* (2016) by Crytek,^[87] *Gigantic* (2017) by Perfect World Entertainment,^[88] *Master X Master* (2018) by NCSoft,^[89] and *Paragon* (2018) by Epic Games.^[90]

The saturation of the market and the dominance of established titles contributed to the decline of many new MOBA attempts, as they struggled to retain player engagement and compete with the genre's leading games.

Impact

In the original *Defense of the Ancients* (*DotA*), each player controls one powerful unit rather than a large army. While it still kept the large scale, core mechanics, and goals of the real-time strategy games, *DotA* attempted to avoid "clickfest" gameplay in which high actions per minute scores are mandatory for efficient playing, changing focus to the actual teamwork, coordination, and tactics. This made the mod highly popular, as its dynamic and unpredictable fights, complex map, and hero-centric gameplay create a more competitive environment and opportunities for outplaying the enemy team.^[21] Over time, the multiplayer online battle arena genre grew steadily within esports tournaments, becoming a major part of the competitive gaming scene by the early 2010s. The genre has seen further growth in popularity since the year 2015 – among the top five esports with the largest prize pools, three have been MOBA titles for three years in row. Distributed prize money in MOBA tournaments reached over US\$54 million in 2017.^[91] A year later, prize pools continued to grow reaching over US\$60 million, 40% of the year's total esports prize pools.^{[92][93]} *2018 League of Legends World Championship* had the biggest prize pool out of all League of Legends esports championship finals, awarding almost \$6.5 million.^{[94][95]}



The *2017 League of Legends World Championship Finals* held in the Beijing National Stadium, China

MOBAs are some of the most watched games in the world.^[96] Major esports professional tournaments are held in venues that can hold tens of thousands of spectators and are streamed online to millions more.^{[97][98][99]} A strong fanbase has opened up the opportunity for sponsorship and advertising, eventually leading the genre to become a global cultural phenomenon.^{[55][100]}

A free-to-play business model, which is used by the largest MOBA titles, have contributed to the genre's overall popularity. Players are able to download and play AAA-quality games at no cost. These games are generating revenue by selling cosmetic elements, including skins, voice lines, customized mounts and announcers, but none of these give the functional gameplay advantages to the buyer. As of 2012, free-to-play MOBAs, such as *League of Legends*, *Dota 2*, *Heroes of the Storm*, and *Smite* were among the most popular PC games.^{[101][102]} The success in the genre has helped convince many video game publishers to copy the free-to-play MOBA model.^{[103][104]} SuperData Research reported that the genre generated over \$2.5 billion of revenue in 2017.^{[105][106]}

Similar to fighting games, MOBAs offer a large number of viable player characters, each of which having distinctive abilities, strengths, and weaknesses. With numerous choices available, players can find the character that best fits their skills and preferences.^[21] Playable characters blend a variety of fantasy

tropes and often reference popular culture and mythology.^{[107][108][109]} One such figure commonly represented in MOBAs is Sun Wukong, a legendary mythical figure from 16th-century China. Examples of representation of Sun Wukong in MOBAs in the form of playable characters include "Wukong" in *League of Legends*, Samuro's "Monkey King" skin (custom outfit/costume) in *Heroes of the Storm*, and "Monkey King" in *DotA 2*.^{[110][111][112]}

Data analytics and match prediction

Due to the large volume of matches played on a daily basis around the world and the relatively complicated nature of the genre,^[c] MOBAs have become a popular target for the application of big data tools to predict match outcomes based on in-game factors such as hero kill/death/assist ratios, gold earned, time of a match, synergy with other players, team composition, and other, more advanced parameters.^{[115][116]}

Artificial intelligence in MOBAs

The use of artificial intelligence in MOBAs is an ongoing topic of research.^[117] Similar to real-time strategy games, MOBAs provide a highly complex environment for AI because of their large amount of possible variables, states, and decisions. One of the first known research-based MOBA AI agents was published around 2015 for *League of Legends*.^[118] The agent used influence maps to navigate the map and compute positioning risk. A similar agent to assist players was published in the same year.^[119]

Two years later, artificial intelligence research laboratory OpenAI developed the AI project OpenAI Five, which was first showcased at the *Dota 2 World Championship, The International 2017*, during a 1v1 demonstration.^[120] In this demonstration, OpenAI Five faced off against Dendi, a *DotA* player. In this 1v1, OpenAI Five defeated Dendi twice in a resounding manner, with the first victory occurring before the five-minute mark and Dendi conceding before ninety seconds had passed in the second match.^[121] OpenAI returned to *The International 2018*, this time fielding an entire team of five AI players in two games against professional players, but ultimately losing both games.^[122] Despite this loss, OpenAI continued to work to improve OpenAI Five, and the project's advancement soon became evident: one year later, at *The International 2019*, OpenAI Five defeated *The International 2018*-winner OG in a limited version of *Dota 2*, becoming the first artificial intelligence system to beat the reigning world champion team at a video game.^[123]

See also

- List of multiplayer online battle arena games

Notes

- Also known as **action real-time strategy (ARTS)** or, more recently, as **hero brawler** and **team brawler**
- The number in brackets represents the year of official servers shut down.
- League of Legends* alone had a reported 100 million active monthly players worldwide in 2016^[113] and an average of 27 million *League of Legends* games played per day reported in 2014.^[114]

References

1. Cannizzo, Alejandro; Ramírez, Esmitt (2015). "Towards Procedural Map and Character Generation for the MOBA Game Genre" (<https://doi.org/10.17230%2Fingciencia.11.22.5>). *Ingeniería y Ciencia*. **11** (22): 95–119. doi:10.17230/ingciencia.11.22.5 (<https://doi.org/10.17230%2Fingciencia.11.22.5>). hdl:10784/7884 (<https://hdl.handle.net/10784%2F7884>). ISSN 1794-9165 (<https://search.worldcat.org/issn/1794-9165>).
2. "Frequently Asked Questions" (<https://web.archive.org/web/20101111183559/http://www.getdota.com/faq>). GetDota.com. Archived from the original (<http://www.getdota.com/faq>) on 11 November 2010.
3. Funk, John (2 September 2013). "MOBA, DOTA, ARTS: A brief introduction to gaming's biggest, most impenetrable genre" (<https://www.polygon.com/2013/9/2/4672920/moba-dota-arts-a-brief-introduction-to-gamings-biggest-most>). *Polygon*. Retrieved 24 November 2019.
4. Walbridge, Michael (12 June 2008). "Analysis: Defense of the Ancients - An Underground Revolution" (https://web.archive.org/web/20120510135818/http://www.gamasutra.com/view/news/109814/Analysis_Defense_of_the_Ancients_An_Underground_Revolution.php). *www.gamasutra.com*. Archived from the original (https://www.gamasutra.com/view/news/109814/Analysis_Defense_of_the_Ancients_An_Underground_Revolution.php) on 10 May 2012. Retrieved 5 September 2020.
5. "How Warcraft 3's modding community paved the way for League of Legends and Dota 2" (<https://www.pcgamesn.com/warcraft-iii/warcraft-3-mods-dota-league-of-legends>). *PCGamesN*. 26 March 2018. Retrieved 6 September 2020.
6. Amstrup, Johannes; ersen (15 September 2017). "Best Modern MOBA Games – LoL, Dota 2, HotS & Smite Compared" (<https://www.progamerreview.com/best-moba-games/>). *Pro Gamer Reviews*. Retrieved 19 October 2019.
7. Huda, Irfan Nabil; Salehudin, Imam (26 March 2024). "Escapism, family and friends, and social interactions: How watching esports increases gaming and in-game purchase intentions" (<https://doi.org/10.25105%2Fjmpj.v17i1.18463>). *Jurnal Manajemen Dan Pemasaran Jasa*. **17** (1): 1–22. doi:10.25105/jmpj.v17i1.18463 (<https://doi.org/10.25105%2Fjmpj.v17i1.18463>).
8. "How MOBAs Took Over Gaming" (<https://me.ign.com/en/pc/70142/feature/how-mobas-took-over-gaming>). *IGN Middle East*. 1 August 2013. Retrieved 5 September 2020.
9. "The Crystal Scar | League of Legends" (<https://web.archive.org/web/20140708150148/http://dominion.leagueoflegends.com/overview-dominion>). *Dominion.leagueoflegends.com*. Archived from the original (<http://dominion.leagueoflegends.com/overview-dominion>) on 8 July 2014. Retrieved 23 December 2015.
10. Leahy, Brian (13 October 2010). "Dota Explained and How You Can Play it Now" (<http://www.shacknews.com/onearticle.x/66007>). *Shacknews.com*. Archived (<https://web.archive.org/web/20101115130020/http://www.shacknews.com/onearticle.x/66007>) from the original on 15 November 2010.
11. "Esports primer: League of Legends" (<https://web.archive.org/web/20210310080830/https://www.reuters.com/article/esports-lol-primer-idUSFLM1tj2jd>). *WebArchive*. Archived from the original (<https://www.reuters.com/article/esports-lol-primer-idUSFLM1tj2jd>) on 10 March 2021. Retrieved 4 June 2025.
12. "Proving Grounds: The Geography of the MOBA Map" (<https://killscreen.com/themeta/proving-grounds-geography-moba-map/>). *The Meta*. 22 September 2016. Retrieved 6 September 2020.
13. Gies, Arthur (2 August 2017). "The Normal Person's Guide to Watching Competitive Dota 2 (2017 Edition)" (<https://www.polygon.com/2017/8/2/16073588/the-normal-persons-guide-to-watching-competitive-dota-2-2017-edition>). *Polygon*. Retrieved 7 September 2020.

14. "Roshan Dota 2 Guide" (<https://web.archive.org/web/20190925110057/https://pages.firstblood.io/pages/blog/dota-2/roshan-dota-2-guide/>). *FirstBlood®*. 17 October 2018. Archived from the original (<https://pages.firstblood.io/pages/blog/dota-2/roshan-dota-2-guide/>) on 25 September 2019. Retrieved 25 September 2019.
15. "Heroes of the Storm: How to Fully Utilize Boss and Mercenary Camps – Articles – Dignitas" (<https://web.archive.org/web/20171030105304/http://team-dignitas.net/articles/blogs/Heroes-Of-The-Storm/11768/heroes-of-the-storm-how-to-fully-utilize-mercenary-camps>). *team-dignitas.net*. 27 October 2017. Archived from the original on 30 October 2017. Retrieved 25 September 2019.
16. "Where should we place the mini map?" (https://web.archive.org/web/20130124090356/http://gamasutra.com/blogs/JacekSliwinski/20130121/185119/Where_should_we_place_the_mini_map.php). *www.gamasutra.com*. 21 January 2013. Archived from the original (https://www.gamasutra.com/blogs/JacekSliwinski/20130121/185119/Where_should_we_place_the_mini_map.php) on 24 January 2013. Retrieved 5 September 2020.
17. "Basic Survival – Learn Dota" (<http://www.playdota.com/learn/survival>). *PlayDota.com*. Archived (<https://web.archive.org/web/20120902023750/http://www.playdota.com/learn/survival>) from the original on 2 September 2012. Retrieved 28 October 2010.
18. "Anniversary Mural & Gifts" (<https://nexus.leagueoflegends.com/en-us/2019/10/anniversary-mural-gifts/>). *nexus.leagueoflegends.com*. 17 October 2019. Retrieved 7 September 2020.
19. Augustine, Rob (3 July 2012). "Introducing League of Legends' 100th champion: Jayce, the Defender of Tomorrow" (<http://www.pcgamer.com/introducing-league-of-legends-100th-champion-jayce-the-defender-of-tomorrow/>). *PC Gamer*. Archived (<https://web.archive.org/web/20161109152828/http://www.pcgamer.com/introducing-league-of-legends-100th-champion-jayce-the-defender-of-tomorrow/>) from the original on 9 November 2016.
20. "Yone announced as League's 150th champion: Abilities, release date, more" (<https://www.dexerto.com/league-of-legends/yone-announced-as-leagues-150th-champion-abilities-release-date-more-1396725/>). *Dexerto*. 24 July 2020. Retrieved 26 January 2022.
21. Crider, Michael (6 November 2017). "Why Are MOBA Games like League of Legends So Popular?" (<https://www.howtogeek.com/331389/why-are-moba-games-like-league-of-legends-so-popular/>). *How-To Geek*. Retrieved 19 July 2020.
22. "Master the hot new mobile game Arena of Valor with our tips" (<https://www.redbull.com/int-en/arena-of-valor-tips>). *Red Bull*. 30 August 2017. Retrieved 5 September 2020.
23. "Why MOBA Games Force Players to Work Together: The Hero Limitation Design Philosophy" (<https://gogamingkits.com/why-moba-games-force-players-to-work-together/>). *gogamingkits.com*.
24. Silva, Victor do Nascimento; Chaimowicz, Luiz (30 May 2017). "MOBA: a New Arena for Game AI". *arXiv:1705.10443* (<https://arxiv.org/abs/1705.10443>) [cs (<https://arxiv.org/archive/cs>)].
25. Biessener, Adam (13 October 2010). "Valve's New Game Announced, Detailed: Dota 2" (<http://gameinformer.com/b/features/archive/2010/10/13/dota-2-announced-details.aspx>). *Game Informer*. Archived (<https://web.archive.org/web/20101016155010/http://gameinformer.com/b/features/archive/2010/10/13/dota-2-announced-details.aspx>) from the original on 16 October 2010. Retrieved 20 October 2010.
26. "League of Legends Roles and Champion Positions" (<https://www.mobachampion.com/blog/league-of-legends-slang-terms-and-meanings/>). *MOBA Champion*. 20 March 2020. Retrieved 5 September 2020.
27. "Heroes of the Storm: Hero Roles Expanded" (<https://www.player.one/heroes-storm-patch-hero-roles-expanded-125044>). *Player.One*. 27 March 2019. Retrieved 5 September 2020.

28. Gavrilova, Nuangjumnong (22 January 2016). "The Effects of Gameplay on Leadership Behaviors: An Empirical Study on Leadership Behaviors and Roles in Multiplayer Online Battle Arena Games" (<https://books.google.com/books?id=4s9yCwAAQBAJ&q=The+Effects+of+Gameplay+on+Leadership+Behaviors%3A+An+Empirical+Study+on+Leadership+Behaviors+and+Roles+in+Multiplayer+Online+Battle+Arena+Games&pg=PA142>). *Transactions on Computational Science XXVI: Special Issue on Cyberworlds and Cybersecurity*. Springer. p. 147. ISBN 9783662492475.
29. Ng, Patrick; Nesbitt, Keith; Blackmore, Karen (5 February 2015). "Sound Improves Player Performance in a Multiplayer Online Battle Arena Game". *Artificial Life and Computational Intelligence*. Lecture Notes in Computer Science. Vol. 8955. Springer, Cham. pp. 166–174. doi:10.1007/978-3-319-14803-8_13 (https://doi.org/10.1007%2F978-3-319-14803-8_13). ISBN 9783319148021.
30. "Understanding Roles in HotS – Articles – Dignitas" (<https://dignitas.gg/articles/blogs/Heroes-Of-The-Storm/13444/understanding-roles-in-hots>). *dignitas.gg*. 27 February 2019. Retrieved 24 September 2019.
31. "The Lost Vikings – Heroes of the Storm" (<http://us.battle.net/heroes/en/heroes/the-lost-vikings/>). *us.battle.net*. Retrieved 24 September 2019.
32. "The Next Generation 1996 Lexicon A to Z". *Next Generation*. No. 15. Imagine Media. March 1996. pp. 28–42. "Action game - A game characterized by simple action and response gameplay. ... the defining characteristic is that enemies and obstacles are overcome by 'physical' means, rather than involved intellectual problem solving."
33. Adams, Rollings 2006
34. PicklePantsLOL (30 June 2020). "How to Hit Skill Shots more Effectively | Day 25/30" (<http://medium.com/@PicklePantsLoL/how-to-hit-skill-shots-more-effectively-d1dbbce69815>). *Medium*. Retrieved 8 September 2020.
35. Adams, Rollings 2003, p. 347
36. "What you need to know about the experience in Dota 2 written by Artem Uarabei | Click-Storm" (<https://click-storm.com/en/articles/6795/>). *Click-Storm.com*. Retrieved 25 September 2019.
37. Jacobacci, Kara (22 April 2016). "Mana - Less Heroes in the ARTS MOBA Genre" (<https://esportsedition.com/dota-2/mana-less-heroes-in-the-arts-moba-genre/>). *Esports Edition*. Retrieved 8 September 2020.
38. "Items | League of Legends" (<https://na.leagueoflegends.com/en/game-info/items/>). *na.leagueoflegends.com*. Retrieved 25 September 2019.
39. "The 5 Most Important Heroes of the Storm Objectives" (<https://www.esportstalk.com/blog/5-most-important-heroes-of-the-storm-objectives-10213/>). *EsportsTalk.com*. 21 November 2018. Retrieved 25 September 2019.
40. *GameAxis Unwired*, p. 52 (<https://books.google.com/books?id=vuoDAAAAMBAJ&dq=herzog+dota&pg=PA52>), December 2008, *SPH Magazines*, ISSN 0219-872X
41. Greg Lockley (3 June 2014), MOBA: The story so far (<http://www.mcvuk.com/news/read/moba-the-story-so-far/0133335>) Archived (<https://web.archive.org/web/20170427151821/http://www.mcvuk.com/news/read/moba-the-story-so-far/0133335>) 27 April 2017 at the *Wayback Machine*, MCV
42. Brown, Fraser (15 January 2013). "Like Macross without the drama" (<http://www.destructoid.com/review-airmech-242461.phtml>). Archived (<https://web.archive.org/web/20130119043712/http://www.destructoid.com/review-airmech-242461.phtml>) from the original on 19 January 2013. Retrieved 19 January 2013.
43. Andrew Groen (7 March 2012), Ask GR Anything: What's a MOBA? (<http://www.gamesradar.com/ask-gr-anything-whats-moba/>) Archived (<https://web.archive.org/web/20170630043636/http://www.gamesradar.com/ask-gr-anything-whats-moba/>) 30 June 2017 at the *Wayback Machine*, *GamesRadar*

44. "Herzog Zwei (or: How I Learned To Stop Worrying And Love Console Multiplayer)" (<https://venturebeat.com/2009/05/08/herzog-zwei-or-how-i-learned-to-stop-worrying-and-love-console-multiplayer/>). *VentureBeat*. 8 May 2009. Retrieved 20 September 2020.
45. "The Essential 50 Part 31: Herzog Zwei" (<https://web.archive.org/web/20120614033131/http://www.1up.com/features/essential-50-herzog-zwei>). *1Up Network*. 4 June 2012. Archived from the original (<http://www.1up.com/features/essential-50-herzog-zwei>) on 14 June 2012. Retrieved 20 September 2020 – via *Internet Archive*.
46. "The Origin of Realtime Strategy Games on the PC" (<http://above-the-garage.com/rblts/vie16b.htm>). 22 November 2011. Archived (<https://web.archive.org/web/20111122065508/http://above-the-garage.com/rblts/vie16b.htm>) from the original on 22 November 2011. Retrieved 20 September 2020.
47. "Future Cop: LAPD" (<http://ign.com/articles/1998/12/15/future-cop-lapd>). *IGN*. 14 December 1998. Archived (<https://web.archive.org/web/20151224105144/http://www.ign.com/articles/1998/12/15/future-cop-lapd>) from the original on 24 December 2015. Retrieved 23 December 2015.
48. "Future Cop L.A.P.D. Review" (<http://www.gamerevolution.com/review/future-cop-lapd>). *Gamerevolution.com*. 1 September 1998. Archived (<https://web.archive.org/web/20151224104032/http://www.gamerevolution.com/review/future-cop-lapd>) from the original on 24 December 2015. Retrieved 23 December 2015.
49. Lopuszanski, Stefan (2019). "It's Mods All The Way Down". *ResearchGate*. doi:10.13140/RG.2.2.25818.08644 (<https://doi.org/10.13140%2FRG.2.2.25818.08644>).
50. "Herzog Zwei" (<https://classicreload.com/genesis-herzog-zwei.html>). *ClassicReload.com*. 15 June 2017. Retrieved 20 September 2020. "The multiplayer mode is only possible via a split-screen setup, in which the competing players can observe the other's activity, making discreet maneuvers or surprise attacks difficult to execute."
51. "Future Cop: LAPD" (<https://www.play-old-pc-games.com/2014/05/23/future-cop-lapd/>). *Play Old PC Games*. 23 May 2014. Retrieved 17 January 2019.
52. Dean, Paul (16 August 2011). "The Story of DOTA" (<http://www.eurogamer.net/articles/2011-08-16-the-story-of-dota-article>). *Eurogamer*. Gamer Network. Archived (<https://web.archive.org/web/20161108042709/http://www.eurogamer.net/articles/2011-08-16-the-story-of-dota-article>) from the original on 8 November 2016. Retrieved 6 April 2014.
53. Dimirti (22 July 2013). "Dota 2: A History Lesson" (<https://web.archive.org/web/20131211054041/http://themittani.com/features/dota-2-history-lesson?page=0%2C0>). The Mittani. Archived from the original (<http://themittani.com/features/dota-2-history-lesson?page=0%2C0>) on 11 December 2013. Retrieved 24 August 2013.
54. Nguyen, Thierry (1 September 2009). "Clash of The DOTAs" (<https://web.archive.org/web/20111128151953/http://www.1up.com/features/clash-dotas-league-legends-heroes>). *1UP.com*. Archived from the original (<http://www.1up.com/do/feature?cld=3175835>) on 28 November 2011. Retrieved 21 October 2009.
55. "The history of MOBAs: From mod to sensation" (<https://venturebeat.com/2014/09/01/the-history-of-mobas-from-mod-to-sensation/>). *VentureBeat*. 1 September 2014. Retrieved 9 September 2020.
56. Crecente, Brian. "League of Legends is now 10 years old. This is the story of its birth" (<https://www.washingtonpost.com/video-games/2019/10/27/league-legends-is-now-years-old-this-is-story-its-birth/>). *Washington Post*. ISSN 0190-8286 (<https://search.worldcat.org/issn/0190-8286>). Retrieved 4 September 2020.
57. Nemikan (21 September 2009). "DOTA reborn: Three games inspired by the legendary WC3 mod" (<https://web.archive.org/web/20120826010826/http://icrantic.com/article/dota-reborn-three-games-inspired-by-the-legendary-wc3-mod>). *Icrantic.com*. Archived from the original (<http://gaming.icrantic.com/reviews/dota-reborn-three-games-inspired-by-the-legendary-wc3-mod/>) on 26 August 2012. Retrieved 17 November 2010.

58. Psychotronic (30 November 2008). "Minions" (<https://web.archive.org/web/20121107114907/http://jayisgames.com/archives/2008/11/minions.php>). jayisgames.com. Archived from the original (<http://jayisgames.com/archives/2008/11/minions.php>) on 7 November 2012. Retrieved 7 November 2012.
59. Lopez, Miguel (21 February 2008). "Demigod" (<https://web.archive.org/web/20120907175611/http://pc.gamespy.com/pc/gas-powered-games-project/854109p1.html>). Gamespy.com. Archived from the original (<http://pc.gamespy.com/pc/gas-powered-games-project/854109p1.html>) on 7 September 2012. Retrieved 20 November 2010.
60. Perez, Daniel (16 January 2009). "Previews; We talk with new developer Riot Games about its strategy/RPG hybrid and the lead designer's Warcraft roots" (<https://web.archive.org/web/20111130044838/http://www.1up.com/previews/league-of-legends>). 1UP.com. Archived from the original (<http://www.1up.com/do/previewPage?cld=3172308&p=1>) on 30 November 2011. Retrieved 7 February 2009.
61. Arirang (3 October 2009). "A Look at the Future of Dota and the AoS Genre" (<https://web.archive.org/web/20120826034014/http://www.gameriot.com/offline/>). GameRiot.com. Archived from the original (<http://wowriot.gameriot.com/blogs/The-Ohcee/DOTA-A-Look-at-the-Future-of-Dota-and-the-AoS-Genre>) on 26 August 2012. Retrieved 19 October 2010.
62. Nutt, Christian (29 August 2011). "The Valve Way: Gabe Newell And Erik Johnson Speak" (<https://www.gamedeveloper.com/design/the-valve-way-gabe-newell-and-erik-johnson-speak>). *Game Developer*. Archived (https://web.archive.org/web/20121023063739/http://www.gamasutra.com/view/feature/6471/the_valve_way_gabe_newell_and_.php?page=3) from the original on 23 October 2012. Retrieved 24 August 2013.
63. Jackson, Leah (23 December 2010). "Looking Back at 2010: The Year in PC Games" (<https://web.archive.org/web/20130113033819/http://www.g4tv.com/thefeed/blog/post/709363/looking-back-at-2010-the-year-in-pc-games/>). g4tv.com. Archived from the original (<http://g4tv.com/thefeed/blog/post/709363/Looking-Back-at-2010-The-Year-in-PC-Games-.html>) on 13 January 2013. Retrieved 24 December 2010.
64. Wedel, Mark (24 June 2010). "Kalamazoo-made 'Heroes of Newerth' drawing huge online gaming crowd" (http://www.mlive.com/entertainment/kalamazoo/index.ssf/2010/06/kalamazoo-made_heroes_of_newer.html). Kalamazoo Gazette. Archived (https://web.archive.org/web/20100628060136/http://www.mlive.com/entertainment/kalamazoo/index.ssf/2010/06/kalamazoo-made_heroes_of_newer.html) from the original on 28 June 2010.
65. "Valve Announces Dota 2" (<http://store.steampowered.com/news/4487/>). Valve. 19 October 2010. Archived (<https://web.archive.org/web/20101015030731/http://store.steampowered.com/news/4487/>) from the original on 15 October 2010.
66. Totillo, Stephen (13 October 2010). "Valve's New Game Is Dota 2" (<https://kotaku.com/valves-new-game-is-dota-2-5663018>). Kotaku. Archived (<https://web.archive.org/web/20121015195518/http://kotaku.com/5663018/valves-new-game-is-dota2/>) from the original on 15 October 2012. Retrieved 17 October 2010.
67. Josh Augustine (17 August 2010). "Riot Games' dev counter-files "DotA" trademark" (<http://www.pcgamer.com/2010/08/17/riot-games-dev-counter-files-dota-trademark/>). PC Gamer. Archived (<https://web.archive.org/web/20130203103306/http://www.pcgamer.com/2010/08/17/riot-games-dev-counter-files-dota-trademark/>) from the original on 3 February 2013. Retrieved 23 December 2015.
68. "Blizzard DotA – BlizzCon 2010 – Matt Gotcher, StarCraft II Level Designer" (<https://www.youtube.com/watch?v=AAgUx0nmgRc>). YouTube.com. 23 October 2010. Archived (<https://ghostarchive.org/varchive/youtube/20211221/AAgUx0nmgRc>) from the original on 21 December 2021. Retrieved 2 March 2013.
69. Augustine, Josh (23 October 2010). "The first heroes in SC2's DOTA map" (<http://www.pcgamer.com/2010/10/23/the-first-heroes-in-sc2s-dota-map/>). PCGamer. Archived (<https://web.archive.org/web/20101024205619/http://www.pcgamer.com/2010/10/23/the-first-heroes-in-sc2s-dota-map/>) from the original on 24 October 2010.

70. Reilly, Jim (11 May 2012). "Valve, Blizzard Reach DOTA Trademark Agreement" (<https://www.gameinformer.com/b/news/archive/2012/05/11/valve-blizzard-reach-dota-trademark-agreement.aspx>). Game Informer. Archived (<https://web.archive.org/web/20120724090129/http://www.gameinformer.com/b/news/archive/2012/05/11/valve-blizzard-reach-dota-trademark-agreement.aspx>) from the original on 24 July 2012.
71. Bramblet, Matthew (1 August 2013). "Diablo III Announcement Coming at Gamescon – Activision Blizzard Q2 2013 earnings report details the Blizzard All-Star progress and 'Project Titan' revamp" (<https://web.archive.org/web/20130809012024/http://diablo.somepage.com/news/1603-diablo-iii-announcement-coming-at-gamescon>). Diablo Somepage. Archived from the original (<http://diablo.somepage.com/news/1603-diablo-iii-announcement-coming-at-gamescon>) on 9 August 2013. Retrieved 19 October 2013.
72. "Blizzard's Diablo/Starcraft/WoW Crossover Has a New Name" (<https://kotaku.com/blizzard-s-diablo-starcraft-wow-crossover-has-a-new-nam-1447238984>). Kotaku. 17 October 2013. Retrieved 7 June 2020.
73. "MOBA Title SMITE Has A Release Date, Coming This March" (<https://www.dsogaming.com/news/moba-title-smite-has-a-release-date-coming-this-march/>). 26 November 2013.
74. "Heroes Of The Storm Proves That A New Warcraft Strategy Game Could Work" (<https://kotaku.com/heroes-of-the-storm-is-proof-a-new-warcraft-strategy-ga-1714960298>). Kotaku. 30 June 2015. Retrieved 9 September 2020.
75. "From Warcraft III to Heroes of the Storm, Talking Art and Blizzard's Long History with Samwise Didier - AusGamers.com" (<https://www.ausgamers.com/features/read/3564332>). *www.ausgamers.com*. Retrieved 24 November 2019.
76. "Blizzard explains why it doesn't call Heroes of the Storm a MOBA" (<http://www.gamespot.com/articles/blizzard-explains-why-it-doesn-t-call-heroes-of-the-storm-a-moba/1100-6416084/>). GameSpot. 9 November 2013. Archived (<https://web.archive.org/web/20131116030735/http://www.gamespot.com/articles/blizzard-explains-why-it-doesn-t-call-heroes-of-the-storm-a-moba/1100-6416084/>) from the original on 16 November 2013. Retrieved 15 November 2013.
77. "Trends In Online Gaming In 2017 And Beyond" (<https://plarium.com/en/blog/trends-online-gaming-2017/>). *plarium.com*. Retrieved 4 July 2020.
78. Webster, Andrew (18 December 2017). "Tencent is bringing China's biggest game to the rest of the world" (<https://www.theverge.com/2017/12/18/16469736/tencent-arena-of-valor-honor-of-kings-north-america-launch>). *The Verge*. Retrieved 24 September 2019.
79. "Pokémon Unite is coming to iOS and Android on September 22" (<https://techcrunch.com/2021/08/18/pokemon-unite-is-coming-to-ios-and-android-on-september-22/>). *TechCrunch*. 18 August 2021. Retrieved 20 August 2021.
80. "Battlerite Review: All the teamfights you'd ever want in a MOBA, minus the MOBA structure" (<https://ag.hyperxgaming.com/article/3709/battlerite-review---moba-battles-done-right>). *AllGamers*. 12 January 2018. Retrieved 26 September 2019.
81. "Paragon preview: Epic Games re-energises the MOBA with classic shooter elements and unique twists" (<https://www.ibtimes.co.uk/paragon-preview-epic-games-re-energises-moba-classic-shooter-elements-unique-twists-1544568>). *International Business Times UK*. 18 February 2016. Retrieved 24 September 2019.
82. Chalk, Andy (27 January 2016). "NCsoft announces Master X Master, a tag-team 'action MOBA'" (<https://www.pcgamer.com/ncsoft-announces-master-x-master-a-tag-team-action-moba/>). *PC Gamer*. Retrieved 24 September 2019.
83. "IGN: Arena of Fate has the 20-minute time limit on each match, and it's all about scoring enough points before that timer runs out" (<https://in.ign.com/preview/4281/what-do-igns-moba-experts-think-of-arena-of-fate>). *IGN India*. 26 June 2014. Retrieved 26 September 2019.
84. "GameSpot: Gigantic is the new free fast-paced hero shooter MOBA" (<https://www.gamespot.com/gigantic/user-reviews/2200-12737495/>). *GameSpot*. Retrieved 26 September 2019.

85. "Dawngate shutting down in 90 days" (<https://www.engadget.com/2014/11/04/dawngate-shutting-down-in-90-days/>). *Engadget*. 4 November 2014. Retrieved 24 September 2019.
86. "Warner Bros. shutting down Infinite Crisis only 2 months after launch" (<https://www.gamesindustry.biz/articles/2015-06-02-warner-bros-shutting-down-infinite-crisis-only-2-months-after-launch>). *GamesIndustry.biz*. 2 June 2015. Retrieved 24 September 2019.
87. "'Crysis' Developer Crytek Shuts Down All But Two Studios" (<https://www.player.one/crisis-developer-crytek-shuts-down-all-but-two-studios-574984>). *Player.One*. 20 December 2016. Retrieved 24 September 2019.
88. McWhertor, Michael (31 January 2018). "Gigantic gets final update, shutting down in July" (<https://www.polygon.com/2018/1/31/16956254/gigantic-shutting-down-january-update-motiga>). *Polygon*. Retrieved 24 September 2019.
89. "NCsoft will shut down Master X Master just months after launch" (<https://www.gamesindustry.biz/articles/2017-11-17-ncsoft-will-shut-down-master-x-master-just-months-after-launch>). *GamesIndustry.biz*. 17 November 2017. Retrieved 24 September 2019.
90. Schreier, Jason (26 January 2018). "After Fortnite's Massive Success, Epic Shuts Down Paragon" (<https://kotaku.com/after-fortnites-massive-success-epic-shuts-down-paragon-1822460782>). *Kotaku*. Retrieved 24 September 2019.
91. "The Top 10 Games of 2017, Based on Distributed Earnings" (<https://esportsobserver.com/top-ten-games-2017/>). *The Esports Observer / home of essential esports business news and insights*. 25 December 2017. Retrieved 22 September 2020.
92. Hurst, Taylor (12 February 2019). "Esports Prize Pools: \$155.9M (2018)" (<https://medium.com/konvoy/esports-prize-pools-155-9m-2018-ce1ff44bd41>). *Medium*. Retrieved 22 April 2020.
93. "The 10 Biggest Esports of 2018 by Total Prize Pool" (<https://esportsobserver.com/10-biggest-esports-prize-pools-2018/>). *The Esports Observer / home of essential esports business news and insights*. 25 December 2018. Retrieved 22 September 2020.
94. "How Many People Play League of Legends & Average LoL Player Stats (2023)" (<https://headphonesaddict.com/league-of-legends-statistics/>). *Headphones Addicts*. 30 November 2022. Retrieved 8 March 2023.
95. "LoL - Esports Viewership and Statistics" (<https://escharts.com/games/lol>). *Esports Charts*. Retrieved 8 March 2023.
96. "Most watched games on Twitch in 2017 – SullyGnome" (<https://sullygnome.com/games/2017/watched>). *sullygnome.com*. Retrieved 18 July 2020.
97. "Video game warriors do battle before 40,000 fans in China" (<https://technology.inquirer.net/68752/technology-games-china-video-games-esports-league-of-legends-birds-nest>). *INQUIRER.net*. Agence France-Presse. 5 November 2017. Retrieved 19 July 2020.
98. Webb, Kevin. "More than 100 million people watched the 'League of Legends' World Championship, cementing its place as the most popular esports" (<https://www.businessinsider.com/league-of-legends-world-championship-100-million-viewers-2019-12>). *Business Insider*. Retrieved 19 July 2020.
99. Boudreau, Ian (26 August 2019). "The International 2019 was Twitch's most-watched Dota 2 event ever" (<https://www.pcgamer.com/the-international-2019-was-twitchs-most-watched-dota-2-event-ever/>). *PC Gamer*. Retrieved 19 July 2020.
100. Meola, Andrew. "The biggest companies sponsoring eSports teams and tournaments" (<https://www.businessinsider.com/top-esports-sponsors-gaming-sponsorships-2018-1>). *Business Insider*. Retrieved 25 September 2020.
101. Gaudiosi, John (11 July 2012). "Riot Games' League Of Legends Officially Becomes Most Played PC Game In The World" (<https://www.forbes.com/sites/johngaudiosi/2012/07/11/riot-games-league-of-legends-officially-becomes-most-played-pc-game-in-the-world>). *Forbes*. Retrieved 5 November 2012.

102. "Q2 2019's Most Impactful PC Videogames: World Cup Boosts Fortnite, Blizzard on the Rise" (<https://esportsobserver.com/q2-2019-impact-index/>). *The Esports Observer* / home of essential esports business news and insights. 27 August 2019. Retrieved 22 September 2020.
103. Drain, Brendan (3 July 2012). "The Soapbox: League of Legends is the new World of Warcraft" (<https://web.archive.org/web/20150202024743/http://massively.joystiq.com/2012/07/03/the-soapbox-league-of-legends-is-the-new-world-of-warcraft>). Joystiq. Archived from the original (<http://massively.joystiq.com/2012/07/03/the-soapbox-league-of-legends-is-the-new-world-of-warcraft>) on 2 February 2015. Retrieved 5 November 2012.
104. Stapleton, Dan (June 1, 2012). "Valve: We Won't Charge for Dota 2 Heroes" (<https://web.archive.org/web/20121030172013/http://pc.gamespy.com/pc/dota-2010/1224922p1.html>). GameSpy. Archived from the original (<http://pc.gamespy.com/pc/dota-2010/1224922p1.html>) on October 30, 2012. Retrieved July 19, 2012.
105. "League of Legends Tops Free-to-Play Revenue Charts in 2017" (<https://comicbook.com/gaming/news/league-of-legends-top-free-to-play-revenue-charts-in-2017/>). *GAMING*. 30 January 2018. Retrieved 18 July 2020.
106. Forde, Matthew; Writer, Staff (16 January 2020). "Moonton's Mobile Legends crosses \$500 million in gross revenue" (<https://www.pocketgamer.biz/asia/news/72366/mobile-legends-500-million-usd-revenue/>). *pocketgamer.biz*. Retrieved 18 July 2020.
107. Marshall, Cass (5 December 2019). "Riot's new games are League of Legends' best asset (and biggest threat)" (<https://www.polygon.com/features/2019/12/5/20992482/riot-games-announcement-lore-storytelling-legends-of-runeterra-league-of-legends>). *Polygon*. Retrieved 7 September 2020. "Now it has lore that's legitimately worth reading, in a collection of genres ranging from Lovecraftian horror to knights and banners fantasy to pirate power struggles."
108. "Blizzard's Worlds Collide When Heroes of the Storm™ Launches June 2" (<https://www.businesswire.com/news/home/20150420005352/en/Blizzard%E2%80%99s-Worlds-Collide-Heroes-Storm%E2%84%A2-Launches-June>). *www.businesswire.com*. 20 April 2015. Retrieved 7 September 2020. "The free-to-play online team brawler brings together a diverse cast of iconic characters from Blizzard's far-flung realms of science fiction and fantasy"
109. "The making of a Smite god: from mythology to main stage" (<https://www.pcgamesn.com/smite/how-smite-s-gods-go-from-mythology-to-main-stage>). *PCGamesN*. 27 November 2015. Retrieved 7 September 2020. "Obviously we're a game about mythology, so one of the things we look for is a mythological character that would be good to bring into the game"
110. "Wukong, the Monkey King - League of Legends" (<https://www.leagueoflegends.com/en-us/champions/wukong/>). *www.leagueoflegends.com*. Retrieved 27 January 2022.
111. S.E.H. @Gnub, Emil. "Monkey King Samuro - Skin" (<https://nexuscompendium.com/skins/samuro-monkeyking>). *The Nexus Compendium*. Retrieved 27 January 2022.
112. "Monkey King" (<https://www.dota2.com/hero/monkeyking>). *Dota 2*. Retrieved 27 January 2022.
113. "Number of players of selected eSports games worldwide as of August 2017 (in million)" (<https://www.riftherald.com/2016/9/13/12865314/monthly-lol-players-2016-active-worldwide>). *Rift Herald*. 13 September 2016. Retrieved 12 August 2018.
114. Sheer, Ian (January 27, 2014). "Player Tally for 'League of Legends' Surges". *Wsj.com*. Archived from the original on January 30, 2014. Retrieved January 31, 2014.
115. Hodge, Victoria; Devlin, Sam; Sephton, Nick; Block, Florian; Drachen, Anders; Cowling, Peter (17 November 2017). "Win Prediction in Esports: Mixed-Rank Match Prediction in Multi-player Online Battle Arena Games". *arXiv:1711.06498* (<https://arxiv.org/abs/1711.06498>) [cs.AI (<https://arxiv.org/archive/cs/AI>)].
116. "LoL Tier List" (<https://www.mobachampion.com/tier-list/>). 17 November 2017. Retrieved 19 December 2018.

117. "OPENAI'S DOTA 2 DEFEAT IS STILL A WIN FOR ARTIFICIAL INTELLIGENCE" (<https://www.theverge.com/2018/8/28/17787610/openai-dota-2-bots-ai-lost-international-reinforcement-learning>). 28 August 2018. Retrieved 19 December 2018.
118. Silva, Victor do Nascimento; Chaimowicz, Luiz (8 June 2017). "On the Development of Intelligent Agents for MOBA Games". arXiv:1706.02789 (<https://arxiv.org/abs/1706.02789>) [cs.AI (<https://arxiv.org/archive/cs.AI>)].
119. Silva, Victor do Nascimento; Chaimowicz, Luiz (9 June 2017). "A Tutor Agent for MOBA Games". arXiv:1706.02832 (<https://arxiv.org/abs/1706.02832>) [cs.AI (<https://arxiv.org/archive/cs.AI>)].
120. Savov, Vlad (14 August 2017). "My favorite game has been invaded by killer AI bots and Elon Musk hype" (<https://www.theverge.com/2017/8/14/16141938/dota-2-openai-bots-elon-musk-artificial-intelligence>). *The Verge*. Retrieved 10 May 2019.
121. Sottek, T. C. (11 August 2017). "The world's best Dota 2 players just got destroyed by a killer AI from Elon Musk's startup" (<https://www.theverge.com/2017/8/11/16137388/dota-2-dendi-open-ai-elon-musk>). *The Verge*. Retrieved 26 January 2022.
122. at 07:04, Katyanna Quach 24 Aug 2018. "Game over, machines: Humans defeat OpenAI bots once again at video games Olympics" (https://www.theregister.co.uk/2018/08/24/openai_bots_eliminated_dota_2/). *www.theregister.co.uk*. Retrieved 10 May 2019.
123. Statt, Nick (13 April 2019). "OpenAI's Dota 2 AI steamrolls world champion e-sports team with back-to-back victories" (<https://www.theverge.com/2019/4/13/18309459/openai-five-dot-a-2-finals-ai-bot-competition-og-e-sports-the-international-champion>). *The Verge*. Retrieved 28 January 2022.

Bibliography

- Adams, Ernest; Rollings, Andrew (2003). *Andrew Rollings and Ernest Adams on game design* (<https://books.google.com/books?id=Qc19ChiOUI4C>). New Riders Publishing. ISBN 978-1-59273-001-8.
- Adams, Ernest; Rollings, Andrew (2006). *Fundamentals of Game Design* (http://wps.prenhall.com/bp_gamedev_1/54/14053/3597646.cw/index.html). Prentice Hall. ISBN 978-0-321-64337-7.

External links

-  Media related to MOBAs at Wikimedia Commons
-

Retrieved from "https://en.wikipedia.org/w/index.php?title=Multiplayer_online_battle_arena&oldid=1350564288"